Proof of key inequality

$$\tilde{\Delta} \log \operatorname{tr}_g \tilde{g} \geq -B \operatorname{tr}_{\tilde{g}} g - \frac{g^{j\bar{k}} \tilde{R}_{j\bar{k}}}{\operatorname{tr}_g \tilde{g}}$$

Compute in normal coordinates at $p \in M$, for g. i.e. at p $g_{j\bar{k}}(p) = \delta_{j\bar{k}}$, $\partial_i g_{j\bar{k}}(p) = 0$. We can assume $\tilde{g}$ is diagonal at p. $\lambda_1, \dots, \lambda_n$ = eigenvalues of $\tilde{g}$ at p.

<u>At p</u>:

$$\tilde{\Delta} \operatorname{tr}_g \tilde{g} = \tilde{g}^{p\bar{q}} \partial_p \partial_{\bar{q}} (g^{j\bar{k}} \tilde{g}_{j\bar{k}})$$

$$= \tilde{g}^{p\bar{q}} (\partial_p \partial_{\bar{q}} g^{j\bar{k}}) \tilde{g}_{j\bar{k}} + \tilde{g}^{p\bar{q}} g^{j\bar{k}} \partial_p \partial_{\bar{q}} \tilde{g}_{j\bar{k}}$$

$$= \tilde{g}^{p\bar{q}} (\partial_p \partial_{\bar{q}} g^{j\bar{k}}) \tilde{g}_{j\bar{k}} - \tilde{g}^{p\bar{q}} g^{j\bar{k}} \tilde{R}_{j\bar{k}p\bar{q}}$$

$$+ \tilde{g}^{p\bar{q}} g^{j\bar{k}} \tilde{g}^{a\bar{b}} (\partial_j \tilde{g}_{p\bar{b}})(\partial_{\bar{k}} \tilde{g}_{a\bar{q}})$$

$$\tilde{g}^{p\bar{q}} (\partial_p \partial_{\bar{q}} g^{j\bar{k}}) \tilde{g}_{j\bar{k}} = \sum_{p,j} \frac{1}{\lambda_p} \lambda_j \underbrace{(\partial_p \partial_{\bar{p}} g^{j\bar{j}})}_{\geq -B} \geq -B \sum_{p,j} \frac{1}{\lambda_p} \lambda_j = -B (\operatorname{tr}_{\tilde{g}} g)(\operatorname{tr}_g \tilde{g})$$

So $$\tilde{\Delta}\, tr_g \tilde{g} \geq -B(tr_{\tilde{g}} g)(tr_g \tilde{g}) - g^{j\bar{k}} \tilde{R}_{j\bar{k}}$$

$$+ \sum_{p,j,a} \frac{1}{\lambda_p \lambda_a} \underbrace{(\partial_j \tilde{g}_{p\bar{a}})(\partial_{\bar{j}} \tilde{g}_{a\bar{p}})}_{|\partial_j \tilde{g}_{p\bar{a}}|^2}$$

$$\tilde{\Delta} \log tr_g \tilde{g} = \tilde{g}^{p\bar{q}} \partial_p \partial_{\bar{q}} (\log tr_g \tilde{g})$$

$$= \tilde{g}^{p\bar{q}} \partial_p \left(\frac{\partial_{\bar{q}} tr_g \tilde{g}}{tr_g \tilde{g}} \right) = \tilde{g}^{p\bar{q}} \frac{\partial_p \partial_{\bar{q}} tr_g \tilde{g}}{tr_g \tilde{g}} - \tilde{g}^{p\bar{q}} \frac{(\partial_p tr_g \tilde{g})(\partial_{\bar{q}} tr_g \tilde{g})}{(tr_g \tilde{g})^2}$$

$$\geq -B\, tr_{\tilde{g}} g - \frac{g^{j\bar{k}} \tilde{R}_{j\bar{k}}}{tr_g \tilde{g}} + \frac{1}{tr_g \tilde{g}} \sum_{p,j,a} \frac{1}{\lambda_p \lambda_a} |\partial_j \tilde{g}_{p\bar{a}}|^2$$

$$- \frac{1}{(tr_g \tilde{g})^2} \sum_{p,a,b} \frac{1}{\lambda_p} (\partial_p \tilde{g}_{a\bar{a}})(\partial_{\bar{p}} \tilde{g}_{b\bar{b}})$$

$$\partial_p tr_g \tilde{g} = \partial_p (g^{j\bar{k}} \tilde{g}_{j\bar{k}})$$

$$= g^{j\bar{k}} \partial_p \tilde{g}_{j\bar{k}} = \partial_p \tilde{g}_{j\bar{j}}$$

Cauchy-Schwarz twice:

$$\sum_{p,a,b} \frac{1}{\lambda_p} (\partial_p \tilde{g}_{a\bar{a}})(\partial_{\bar{p}} \tilde{g}_{b\bar{b}}) = \sum_{a,b} \left(\sum_p \frac{1}{\sqrt{\lambda_p}} \partial_p \tilde{g}_{a\bar{a}} \frac{1}{\sqrt{\lambda_p}} \partial_{\bar{p}} \tilde{g}_{b\bar{b}} \right)$$

$$\leq \sum_{a,b} \left(\sum_p \frac{1}{\lambda_p} |\partial_p \tilde{g}_{a\bar{a}}|^2 \right)^{1/2} \left(\sum_q \frac{1}{\lambda_q} |\partial_q \tilde{g}_{b\bar{b}}|^2 \right)^{1/2}$$

$$= \left(\sum_a \left(\sum_p \frac{1}{\lambda_p} |\partial_p \tilde{g}_{a\bar{a}}|^2 \right)^{1/2} \right)^2$$

$$= \left(\sum_a \sqrt{\lambda_a} \left(\sum_p \frac{1}{\lambda_p \lambda_a} |\partial_p \tilde{g}_{a\bar{a}}|^2 \right)^{1/2} \right)^2$$

$$\leq \sum_a \lambda_a \sum_{b,p} \frac{1}{\lambda_p \lambda_b} \underbrace{|\partial_p \tilde{g}_{b\bar{b}}|^2}_{= \partial_b \tilde{g}_{p\bar{b}}}$$

$$\leq (\mathrm{tr}_g \tilde{g}) \sum_{j,b,p} \frac{1}{\lambda_p \lambda_b} |\partial_j \tilde{g}_{p\bar{b}}|^2$$

$$\Longrightarrow \frac{1}{(\mathrm{tr}_g \tilde{g})^2} \sum_{p,a,b} \frac{1}{\lambda_p} (\partial_p \tilde{g}_{a\bar{a}})(\partial_{\bar{p}} \tilde{g}_{b\bar{b}})$$

$$\leq \frac{1}{\mathrm{tr}_g \tilde{g}} \sum_{j,b,p} \frac{1}{\lambda_p \lambda_b} |\partial_j \tilde{g}_{p\bar{b}}|^2$$

so $\tilde{\Delta} \log \operatorname{tr}_g \tilde{g} \geq -B \operatorname{tr}_{\tilde{g}} g - \dfrac{g^{j\bar{k}} \tilde{R}_{j\bar{k}}}{\operatorname{tr}_g \tilde{g}}$. $\square$

So far we know that if $f : M \to \mathbb{R}$ satisfies

$$\left.\begin{aligned} \det(g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f) &= e^F \det(g_{j\bar{k}}) \\ \tilde{g}_{j\bar{k}} = g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f &> 0 \\ \sup_M f &= 0 \end{aligned}\right\}$$

then $\sup_M |f|, \ \sup_M |\partial\bar{\partial} f|_g < C \left(\begin{smallmatrix} \text{dep. on} \\ (M,g), F \end{smallmatrix} \right)$

$\Longrightarrow$ also $C^{-1} g_{j\bar{k}} < \tilde{g}_{j\bar{k}} < C g_{j\bar{k}}$

<u>Step 4</u> Bounding certain 3rd derivatives of f, equivalently $\|\tilde{g}\|_{C^1}$. Better to work with a tensor: let $S^i_{jk} = \tilde{\Gamma}^i_{jk} - \Gamma^i_{jk}$

where $\tilde{\Gamma}, \Gamma$ are Christoffel symbols of $\tilde{g}, g$.

$$\tilde{\Gamma}^i_{jk} = \tilde{g}^{i\bar{l}} \partial_j \tilde{g}_{k\bar{l}}$$

Bounding $|S|_{\tilde{g}}$ is equivalent to bounding $\|\tilde{g}\|_{C^1}$.

<u>Key inequality</u> : There is a constant C (depending on (M,g), F through previous estimates) such that

$$\tilde{\Delta} |S|^2_{\tilde{g}} \geq -C|S|^2_{\tilde{g}} - C \qquad (*)$$

<u>Estimating $|S|^2_{\tilde{g}}$ assuming $(*)$</u> :

From before we have

$$\tilde{\Delta}\, tr_g \tilde{g} \geq -C_1 + \varepsilon |S|^2_{\tilde{g}} \quad \text{for some } C_1, \varepsilon > 0$$

$\Rightarrow \exists$ large $A > 0$ s.t.

$$\tilde{\Delta}\big(\underbrace{|S|^2_{\tilde{g}} + A\, tr_g \tilde{g}}_{h}\big) \geq |S|^2_{\tilde{g}} - C_2 \quad \text{for some } C_2 > 0$$

If $h(p) = \max_M h$, then

$$|S|^2_{\tilde{g}}(p) - C_2 \leq 0$$

$$|S|^2_{\tilde{g}}(p) \leq C_2$$

$$\Rightarrow h(p) \leq C_3$$

$$\Rightarrow h(x) \leq C_3 \text{ for all } x \in M$$

$$\Rightarrow |S|^2_{\tilde{g}} \leq C_3 \text{ on } M. \qquad \square$$